DATA 3464: Fundamentals of Data Processing

Interaction Effects

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Topic overview

- Definitions and description of interaction effects
- Detecting interaction effects
- Feature selection

Resources used:

- Feature Engineering Chapter 7
- Feature Engineering Chapter 10

Definition

> *Two or more predictors are said to interact if their combined effect is different (less or greater) than what we would expect if we were to add the impact of each of their effects when considered alone. -- Feature Engineering, Ch 7*

- Interactions matter if they affect the *outcome*
- Features may have a relationship without having an interaction effect

> *Example: Stroke data from week 1*

Mathematical representation

A linear model is trying to fit:

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_k x_k$$

To consider interaction effects, we add the **product** of features. For a model with only features x_1 and x_2:

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_1 x_2$$

> *This is similar to a **basis expansion** transformation that adds polynomial terms*
> *Scikit-learn's PolynomialFeatures module can do both at once*

Note: common to use β or θ for the weights instead of w, especially in statistics